

DecisionIQ Master Handbook

CHAPTER 6: Core Objectives and Architectural Blueprint of the Platform

6.1 Core Architectural Objectives

DecisionIQ is designed to meet eight core engineering objectives:

DecisionIQ Core Flow

1. Ingestion: Generate 3 years of multi-tenant data

2. Validation: Rigid range & FK audits

3. Storage: SQL schema with primary & foreign keys

4. In-DB Analytics: Perform cohort and RFM segmentations

5. Predictions: ML churn, weekly forecasts & delays

6. Insight Engine: Automate root-cause audits

7. Executive UI: Consulting-grade Streamlit web portal

8. Financial Outcomes: Link recommendations to dollar value

- 1. **Enterprise Ingestion:** Simulate 3 years of multi-tenant data logs across Sales, Finance, Marketing, Logistics, and CS.
- 2. **ETL Quality Control:** Validate data inputs via rigid type and range checks before loading.
- 3. **Star/Snowflake Schema DW:** Build a normalized warehouse schema supporting full PK/FK constraints.
- 4. **Advanced SQL Aggregations:** Calculate cohort retention curves, RFM segmentation, and personnel hierarchies in the database.
- 5. **Active Machine Learning:** Train and serialize models to forecast revenue, predict customer churn, and classify logistics risk.
- 6. **Heuristic Insight Diagnostics:** Auto-trace performance drop indicators to identify root causes and calculate financial leakages.
- 7. **Consulting-Grade UI:** Design a clean Streamlit dashboard utilizing professional design palettes.
- 8. **Actionable Recommendations:** Connect business warnings to prescribed operational actions with calculated financial benefits.